

Formal Methods for Information Security

Spring Semester 2024 ([263-4600-00L](#))

Overview

Lecturers and Tutors:

[Dr. Ralf Sasse](#), [Dr. Christoph Sprenger](#) and [Dr. Srdan Krstic](#)

Lectures:

Tuesday 10-12 [CAB G 59](#)

Exercises:

Tuesday 12-13 [CAB G 59](#)

Credits: 5 ECTS (2V + 1U + 1A)

Project:

25% of the grade

Homework:

optional, but strongly recommended

Exams:

Oral exam (session examination - mid-to-late August), date individually communicated by examination planning office (Prüfungsplanstelle), usually in late June.

Language: English

Announcements

- Welcome the the course. First lecture starts on Tue, 20 Feb at 10h.

Description

The course treats formal methods mainly for the modeling and analysis of security protocols. Cryptographic protocols (such as SSL/TLS, SSH, Kerberos, SAML single-sign on, and IPSec) form the basis for secure communication and business processes. Numerous attacks on published protocols show that the design of cryptographic protocols is extremely error-prone. A rigorous analysis of these protocols is therefore indispensable, and manual analysis is insufficient. The lectures cover the theoretical basis for the (tool-supported) formal modeling and analysis of such protocols. Specifically, we discuss their operational semantics, the formalization of security properties, and techniques and algorithms for their verification.

The second part of this course will cover a selection of advanced topics in security protocols such as abstraction techniques for efficient verification, secure communication with humans, the link between symbolic protocol models and cryptographic models as well as RFID protocols (a staple of the Internet of Things) and electronic voting protocols, including the relevant privacy properties. Moreover, we will give an introduction to two additional topics: non-interference as a general notion of secure systems, both from a semantic and a programming language perspective (type system), and runtime verification/monitoring to detect violations of security policies expressed as trace properties.

Resources

Literature

The lecture is based mainly on various journal/conference papers, but see also (all available at the library):

- Benedikt Schmidt, "[Formal analysis of key exchange protocols and physical protocols ↗](#)", 2012
- Baader and Nipkow, "[Term Rewriting and All That ↗](#)", 1998.
- Cremers and Mauw, "[Operational semantics and verification of security protocols ↗](#)", 2013
- Boyd and Mathuria, "[Protocols for authentication and key establishment ↗](#)", 2003
- Bella, "[Formal correctness of security protocols ↗](#)", 2007
- Ryan and Schneider, "[The modeling and analysis of security protocols ↗](#)", 2001
- Menezes, van Oorschot, and Vanstone, "[Handbook of applied cryptography ↗](#)", 1997
- Basin et al., "[Almost event-rate independent monitoring ↗](#)", 2019
- Basin et al., "[Monitoring Metric First-order Temporal Properties ↗](#)", 2015
- John Rushby, "[Noninterference, transitivity, and channel-control security policies ↗](#)", 1992.

Course Material

The important links, lecture notes, exercises, slides, and other resources are available on [Moodle](#).

Related Content

Formal Methods and Functional Programming

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Information Security

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